

Tests & Diagnostics — AGEL0206 σ_e ApJL paper

Last updated: 2026-05-01 Headline: $\sigma_e(<R_e) = 268 \pm 32$ km/s (stat $\pm 24 \oplus$ I-shape $\pm 13 \oplus$ mask $\pm 16 \oplus$ frame $\pm 5 \oplus$ centering ± 4) Method: §6cum cumulative I-weighted ppxf at $R<R_e=2.305''$ inside the F200LP-masked, frame-aware, SPS-pooled (FSPS+EMILES+XSL) bootstrap $\times$ polynomial-degree posterior

This document is the canonical index of every test, sweep, audit, and sensitivity check run for the σ_e measurement. Each row points to the script/notebook that runs it and the result file or note it produced.

0. Quick-reference test matrix

#	Test	Where	Result	Status
A. Foundational				
A1	KCWI cube provenance	NOTES_methodology_2026-04-27.md	Multi-night Aug/Sept/Dec 2025; header mislabel only	✓
A2	Systemic redshift via line fitting	notebooks/04_redshift_verification.ipynb, scripts/redshift_verify.py	$z = 0.67564$ (vs DR2 0.67511)	✓
A3	R_e from masked curve-of-growth (F140W+F200LP)	scripts/measure_Re.py, scripts/final_sigma_e.py:curve_of_growth	$R_e = 2.305'' = 16.23$ kpc	✓
A4	HST-mean center via centroid_2dg	scripts/final_sigma_e.py:find_center	F140W & F200LP agree to 0.36''	✓
A5	Bagpipes SED fitting ($M_\star$)	notebooks/02_streamlined_Bagpipes_SED.ipynb, 08_sersic_total_photometry.ipynb	$\log M_\star = 11.33 \pm 0.07/-0.09$	✓
B. Pipeline correctness audits				
B1	Instrumental LSF audit (DISPCAL=0.294)	scripts/ifu_spectral_resolution.py	FWHM=0.692 Å; $\sigma_{v_{\text{inst}}} \approx 12\text{-}14$ km/s	✓
B2	σ_{inst} sensitivity ($\times 0.5$ to $\times 2.0$ LSF)	scripts/sigma_inst_sensitivity.py, audit 3	$\max \Delta\sigma = 0.83$ km/s	✓
B3	SPS template wavelength frames	scripts/audit_ppxf_methodology.py, NOTES Test 2	FSPS=vacuum, EMILES=air, XSL=air (Ca K minimum + V_{sys})	✓
B4	V_{sys} air vs vacuum $\times 5$ polynomial degrees	scripts/audit_ppxf_methodology.py audit 1	ΔV consistent within ± 2 km/s across degrees	✓
B5	$z \times$ air-vac differential (obs vs rest application)	scripts/audit_ppxf_methodology.py audit 2	1.83 km/s differential — sub-budget	✓
B6	fwhm_gal_dict frame consistency check	scripts/audit_ppxf_methodology.py audit 4	dict in REST frame, matches Cappellari pattern	✓
C. SPS templates & frame-fix				
C1	Frame-aware ppxf: per-SPS native frame	scripts/bootstrap_ppxf.py:SPS_NATIVE_FRAME + frame_galaxy='auto'	V_{sys} split collapsed from ~ 110 to ~ 15 km/s	✓
C2	End-to-end V_{sys} closure (frame swap)	NOTES Test 3, addendum 2026-04-29	Frame fix matches diagnosis	✓
C3	σ shift from frame fix	NOTES addendum, error_budget()	≤ 5 km/s across SPS $\rightarrow$ carried as ± 5 km/s budget	✓
C4	3-SPS pooled posterior	scripts/final_sigma_e.py:pool_sps	Per-SPS V_{sys} subtracted before pooling	✓
D. Aperture & $I(r)$				
D1	§6cum cumulative I-weighted ppxf	scripts/final_sigma_e.py:run_aperture_sps	$\sigma_e(<R_e) = 267.82$ km/s headline	✓

#	Test	Where	Result	Status
D2	§6cum vs §7 (annular) cross-check	notebooks/07c, ~/.claude/.../reference_cumulative_vs_annular_sigma_e.md	§7=255, §7b=271 — all <1 σ from §6cum	✓
D3	I(r) shape sweep (10 shapes, fixed mask)	scripts/run_isource_shape_sweep.py, scripts/analyze_isource_shape_sweep.py	Range = 12.9 km/s (excluding F200LP_Sersic2D outlier) → ±13 budget	✓
D4	I(r) shape sweep refresh post-frame-fix at N=500	NOTES addendum (d), 2026-04-29	Frame fix has <0.5 km/s impact; ±13 budget validated	✓
D5	Aperture choice: $R < R_e/8$, $R < R_e/2$, $R < R_e$	scripts/final_sigma_e.py: APERTURE_FRACS	$R < R_e/8$ dropped (inside seeing FWHM/2 = 0.64")	✓
D6	Equal-N vs equal-width annular binning	notebooks/07b (5-bin), notebooks/07c (equal-N)	Equal-N inside $R_{\text{safe}} = 3R_e/4$ chosen	✓
D7	R_e source systematic (mean vs F140W vs F200LP vs CaHK+G)	scripts/final_sigma_e.py §7, notebooks/09 §7b	Spread = 16.9 km/s (mean=268, F140W=265, F200LP=276, CaHK=282); sub-budget	✓
E. Mask treatment				
E1	F200LP arc mask reprojected to IFU grid	scripts/final_sigma_e.py: load_setup	0.7% of all spaxels flagged (~38 inside $R < R_e$)	✓
E2	Hard mask (w=0.0) headline	scripts/final_sigma_e.py, track A	$\sigma_e = 267.82 \pm 24$ km/s	✓
E3	No-mask sensitivity (w=1.0)	track B	$\sigma_e = 252.82$, $\Delta = -15.0$ km/s	✓
E4	Soft-mask single-point (w=0.5)	scripts/soft_mask_track.py, track C, NOTES addendum (b)	$\sigma_e = 258.54$, $\Delta = -9.29$; super-linear	✓
E5	5-point mask-weight sweep $w \in \{0, 0.25, 0.5, 0.75, 1\}$	scripts/soft_mask_track.py --weight ..., scripts/analyze_mask_weight_sweep.py, NOTES addendum (c)	Per-step drops 5.0 → 4.3 → 3.5 → 2.2; quadratic c = +7.3 (concave-up); threshold-dominated	✓
E6	Mask dilation (over-masking diagnostic)	notebooks/07d_sigma_e_forceful_mask.ipynb, ~/.claude/.../project_nb07d_overmasking_finding.md	Dilation inflates σ via noise; do NOT dilate (negative finding)	✓
E7	Arc-spectrum subtraction sibling	notebooks/07e_sigma_e_arc_subtract.ipynb	Matches §6cum within 0.1 km/s — residual arc dilution sub-dominant	✓
E8	Arc-as-sky mechanism test (no-mask + arc-sky template)	scripts/run_07f_arc_sky.py, notebooks/07f_arc_sky_subtract.ipynb	$\sigma_D = 274.6$ vs $\sigma_A = 267.8$ / $\sigma_B = 252.8$; recovery=145% — dilution explains the full no-mask Δ	✓
F. Centering				
F1	5-center sweep ($\pm 0.4''$ perturbations)	NOTES_centering_investigation_2026-04-27.md, scripts/regen_s6cum_nomask_diagnostic.py	σ_e spread = 3.7 km/s → ±4 km/s budget	✓
F2	HST F140W vs F200LP centroid offset	scripts/final_sigma_e.py: find_center	0.36" — driven by F200LP arc; HST-mean robust	✓
G. Bootstrap & polynomial degree				
G1	Wild bootstrap (Rademacher $\times$ local residual)	scripts/bootstrap_ppxf.py: run_bootstrap_single_degree	75-pix rolling window for σ_{loc}	✓

#	Test	Where	Result	Status
G2	Parallel bootstrap (joblib, BLAS=1 per worker)	scripts/bootstrap_ppxf_parallel.py	~6 min for 6 fits at N=500	✓
G3	Polynomial degree sweep (deg 15-29)	notebooks/03c_scripts/build_nb09.py:§6.5	σ flat within bootstrap envelope; polynomial saturated	✓
G4	N=50 smoke vs N=500 production agreement	scripts/final_sigma_e.py --n_bootstrap	Within 1 km/s	✓
H	Cross-checks (independent estimators)			
H1	§7 discrete Gültekin annular sum (frame-aware, refresh 2026-05-01)	scripts/refresh_07c_gultekin.py, notebooks/07c §7	256.17 —13.0/+12.7 km/s (post-fix; was 254.99 —24/+28 pre-fix)	✓
H2	§7b flat- σ extrapolation (frame-aware, refresh 2026-05-01)	scripts/refresh_07c_gultekin.py, notebooks/07c §7b	274.37 —16.2/+17.4 km/s (post-fix; was 271 —33/+35 pre-fix)	✓
H3	F200 mask sensitivity at N=500	results/annular_bootstrap_07c_nomask/	$\Delta_{\text{mask}} = -16.4 \text{ km/s} \rightarrow \pm 16 \text{ budget}$	✓
I. Earlier sigma-discrepancy work (Tests 1-21)				
I1	NB01 vs NB05 σ reproduction	notebooks/05x Tests 1-4	Confirmed σ depends on aperture	✓
I2	Source mask + PSF-aware masking	notebooks/05x Tests 7-10	F140W mask flags deflector core; F200LP mask is the right arc mask	✓
I3	Threshold sweep + contamination weighting	notebooks/05x Tests 11-13	Pre-Gültekin masking strategy work	✓
I4	V_rms profile + bootstrap	notebooks/05x Tests 14-16	Pre-Gültekin radial work	✓
I5	Redshift sensitivity (z=0.67511 vs 0.67564)	notebooks/05x Test 17	<1 km/s effect on σ	✓
I6	Voronoi binning attempt	notebooks/05x Test 18	Failed — abandoned	✓
I7	PowerBin spatial binning	notebooks/05x Test 19, scripts/run_powerbin_test19.py	Rejected as binning scheme — outer-bin $\sigma > 800 \text{ km/s}$	✓
I8	R_e four-way comparison	notebooks/05x Test 20, scripts/measure_Re.py	F140W+F200LP masked CoG mean = 2.305" headline	✓
I9	Per-spaxel vs PowerBin rotation map	notebooks/05x Test 21	No rotation on Re-scale; σ -dominated	✓

1. Headline numbers

σ_e at three apertures (cumulative I-weighted ppxf, frame-aware, masked)

Aperture	σ_e [km/s]	Notes
$R < R_e (=2.305")$	268 ± 32 (stat ± 24)	paper headline — KH13/Greene+20 σ_e
$R < R_e/2 (=1.152")$	224 ± 18	gradient diagnostic
$R < R_e/8 (=0.288")$	dropped	inside seeing FWHM/2 = 0.64"

Error-budget composition ($R < R_e$)

Component	Value	Source
Statistical (bootstrap pooled 1σ)	± 24	nb09, this paper

Component	Value	Source
I-shape spread (10 shapes excluding outlier)	± 13	nb07c sweep, refreshed N=500 post-frame-fix
Mask on/off Δ	± 16	nb07c, nb09 §5
Frame fix (max σ shift across SPS)	± 5	NOTES addendum + audit 1
Centering (5-center sweep, $\pm 0.4''$)	± 4	NOTES_centering_investigation_2026-04-27.md
Quadrature total	± 32	scripts/final_sigma_e.py:error_budget

Sensitivity values (for paper text)

- Soft-mask ($w=0.5$) at $R < R_e$: 258.5 km/s — Δ from headline = -9.3 km/s
- No-mask ($w=1.0$) at $R < R_e$: 252.8 km/s — Δ from headline = -15.0 km/s
- $\sigma_e(w)$ is essentially linear with weak super-linearity (quadratic $c=+7.3$)

2. Detailed test catalog

This expands each row of the matrix above. Subsections reference scripts, notebooks, and result files — open the linked file for the detailed code and the inline plots/output.

A. Foundational data and geometry

A1 KCWI cube provenance — The cube Nov17_2025_DESJ0206_RL_combined_icubes_wcs.fits is the final multi-night reduction (Aug 29 K409 + Sept 29 K409 + Dec 29 + Nov 17 2025). The header is mislabeled (DATE-OBS/PROGNAME/PROGPI describe only the last stacking pass). Settled in NOTES_methodology_2026-04-27.md. Cite the multi-night provenance, not the header.

A2 Systemic redshift — Independent line-by-line Gaussian fitting of Ca H+K, [OII], G-band, etc. gives $z_{\text{systemic}} = 0.67564$ (NIST air wavelengths). DR2 catalog $z=0.67511$ (used by older notebooks 03/03b) is offset by ~ 95 km/s from the line-fit value; nb09 uses 0.67564 throughout. Notebook: notebooks/04_redshift_verification.ipynb Module: scripts/redshift_verify.py (provides ABSORPTION_LINES / EMISSION_LINES dicts, gaussian_fit, multi_line_fit, oii_doublet_fit)

**A3 Effective radius $R_e = 2.305''$ — Mean of F140W and F200LP masked curve-of-growth values (2.168 and 2.441''). The masked CoG zeros mask-flagged HST pixels before the radial integral so the arc/companion contributions don't bias the bright-end half-light point. Computed in scripts/measure_Re.py and re-derived in scripts/final_sigma_e.py:curve_of_growth for the nb09 pipeline. Cross-check: IFU white-light proper-masked CoG gave 2.61'' earlier — the HST-based 2.305'' is now the headline.

A4 HST-mean center — photutils.centroid_2dg on F140W and F200LP cutouts (with their own _mask.fits as the mask) gives sub-pixel centroids in each HST WCS. We average in world coordinates, then propagate to IFU sub-pixel: scripts/final_sigma_e.py:find_center. F140W vs F200LP centroid offset = $0.36''$ — flags the F200LP arc-affected detection but the HST-mean is robust (verified by the 5-center sweep, F1).

A5 Stellar mass — Bagpipes SED fitting on HST + JWST aperture photometry $\rightarrow \log M_\star/M_\odot = 11.33 +0.07/-0.09$ (notebook 02). Cross-checked at the Sersic-total level in notebooks/08_sersic_total_photometry.ipynb.

B. Pipeline correctness audits (scripts/audit_ppxf_methodology.py)

Four orthogonal correctness audits run end-to-end on the headline aperture spectrum to verify ppxf usage matches Cappellari (2017, 2023). Saved in results/ppxf_methodology_audit.npz. Documented in NOTES_nb09_frame_fix_and_final_sigma_e_2026-04-28.md ADDENDUM 2026-04-29.

B1 Instrumental LSF — KCWI header DISPSCAL = 0.294 (instrumental σ in pixels) $\times$ CD3_3 = 1.000 Å/pix $\rightarrow$ FWHM_inst = $2.355 \times 0.294 = 0.692$ Å (constant in observed Å). At the rest-frame fit window (3879–4476 Å), $\sigma_{v_inst} \approx 12\text{--}14$ km/s. Galaxy $\sigma \approx 270$ km/s is resolved by $\sim 20\times$. Diagnosed in scripts/ifu_spectral_resolution.py.

B2 σ_{inst} sensitivity — Sweep DISPSCAL $\times$ {0.5, 0.75, 1.0, 1.25, 1.5, 2.0}. Max $|\Delta\sigma| = 0.83$ km/s across all SPS. Reason: ppxf's sps_lib clips $\text{fwhm_diff}^2 = (\text{FWHM_gal}^2 - \text{FWHM_tem}^2).clip(0)$ (sps_util.py:169). When templates are intrinsically broader (FSPS, EMILES at our band), no convolution is applied — σ becomes insensitive to DISPSCAL. LSF is rock-solid. Script: scripts/sigma_inst_sensitivity.py

B3 SPS template frames — Located the Ca K minimum in each shipped template: - FSPS spectra_fsps_9.0.npz: Ca K at 3934.86 Å $\rightarrow +1.20$ Å vs air rest = vacuum - EMILES spectra_emiles_9.0.npz: 3933.82 Å $\rightarrow +0.16$ Å = air - XSL spectra_xsl_9.0.npz: V_sys closes at +9 km/s with air-conversion $\rightarrow$ air End-to-end V_sys closure (NOTES Test 3) confirms.

B4 V_{sys} air vs vacuum $\times 5$ polynomial degrees — Run ppxf at degrees {15, 18, 21, 24, 27} for each SPS in both frames. $\Delta V(\text{air} - \text{vac}) \approx +83$ km/s for FSPS, ≈ -82 km/s for EMILES & XSL, consistent within ± 2 km/s across degrees. Frame identification is robust.

B5 $z \times \text{air-vac}$ differential — Apply `vac_to_air` at observed wavelengths (KCWI 6500–7500 Å) $\rightarrow v_{\text{offset}} = -82.7$ km/s. Apply at rest wavelengths (3879–4476 Å) $\rightarrow -84.5$ km/s. Differential = 1.83 km/s — sub-budget. Either convention is defensible; we follow Cappellari’s pattern (apply at obs).

B6 `fwhm_gal_dict` frame check — `Confirmed_prep_spectrum_for_ppxf` builds the dict as {"lam": `lam_gal_rest`, "fwhm": `fwhm_gal_rest`} in REST frame, matching `ppxf_example_high_redshift.py:99–101`. `FWHM_obs = 0.692` Å $\rightarrow$ `FWHM_rest = 0.413` Å at $z=0.67564$. `sps_util.py:167` interpolates this onto template `lam_temp` (also rest), then `varsmooth` pre-broadens.

C. SPS templates and frame-fix

C1 Frame-aware ppxf — `scripts/bootstrap_ppxf.py` exposes `frame_galaxy='auto'` which reads `SPS_NATIVE_FRAME = {fsp: vacuum, emiles: air, xsl: air}`. The galaxy spectrum is prepared in the matching frame (`util.vac_to_air()` applied as scalar median when needed, matching `ppxf_example_kinematics_sdss.py:127–134`). Diagnosed Apr 2026 — pre-fix the pipeline always converted to air, producing a -90 km/s FSPS V_{sys} offset.

C2 V_{sys} closure — Same galaxy spectrum, swap only the frame. Pre-fix: FSPS -90 , EMILES $+3$, XSL $+9$. Post-fix (frame-aware): FSPS -7 , EMILES $+3$, XSL $+9$. Split-track collapsed from $\sim 110 \rightarrow \sim 15$ km/s. NOTES Test 3.

C3 σ shift from frame fix — Side-effect on σ : ≤ 5 km/s per SPS, ≤ 2 km/s on the 3-SPS pool. Carried as ± 5 km/s in the error budget (ironically dominated by the per-SPS shifts that pool out, but conservative).

C4 3-SPS pooled posterior — `scripts/final_sigma_e.py:pool_sps` concatenates per-SPS bootstrap σ samples after subtracting per-SPS V_{sys} medians (so V offsets don’t inflate the V posterior; σ posteriors are concatenated raw — no V_{sys} correction needed for σ). The headline $\sigma_e(<R_e)$ is the 16/50/84 percentile of this combined-SPS pool.

D. Aperture and I(r) framework

D1 §6cum cumulative I-weighted ppxf — For each spaxel inside $R < R_{\text{max}}$, build I-weight from the IFU 6500–7500 Å white-light band, drop arc-mask-flagged spaxels, normalize and sum to a single 1-D spectrum, fit ppxf. This is the headline path — matches KH13 / Greene+20 / SAURON / ATLAS3D / MaNGA σ_e definition (Cappellari+2006 eq. 1). Cross-references in `~/claude/.../reference_cumulative_vs_annular_sigma_e.md`. Code: `scripts/final_sigma_e.py:run_aperture_sps`.

D2 §6cum vs §7 (annular) — §7 discrete Gültekin annular sum at $R < R_{\text{safe}}$ gives 254.99 km/s; §7b flat- σ extrapolation gives 271 km/s; both within 1σ of §6cum’s 267.82 km/s. §6cum is the headline because (a) it matches the literature definition, (b) it preserves LOSVD line-shape information that §7’s moment-pooling discards, (c) bright-center I-weighting auto-suppresses arc contamination.

D3 I(r) shape sweep (10 shapes) — Hold the spaxel selection fixed (F200-raw mask, $R < R_e$), vary only the I-weight map. Shapes: 1. IFU_band — 6500-7500 Å (headline) 2. IFU_wl — full IFU white-light 3. F140W_raw — HST F140W reprojected 4. F200LP_raw — HST F200LP reprojected 5. F140W_arcmasked — F140W with arc zeroed 6. F200LP_arcmasked — F200LP with own arc-mask zeroed 7. F140W_1Dcog — annular-mean F140W $\rightarrow$ `interp1d` to spaxels 8. F200LP_1Dcog — annular-mean F200LP $\rightarrow$ `interp1d` 9. F140W_Sersic2D — full 2D Sersic fit 10. F200LP_Sersic2D — full 2D Sersic fit (poor fit, $n=0.30$ — outlier)

Range excluding F200LP_Sersic2D outlier = 12.9 km/s $\rightarrow \pm 13$ km/s budget. Script: `scripts/run_isource_shape_sweep.py` Analysis: `scripts/analyze_isource_shape_sweep.py` Caches: `results/annular_bootstrap_07c_ishape/{shape}_{sps}.npz` Figure: `results/figures/nb07c_ishape_sweep.png`

D4 I-shape sweep refresh post-frame-fix — Original sweep at $N=250$ predated the frame fix. Re-ran at $N=500$ with `frame_galaxy='auto'` on 2026-04-29. Per-SPS σ shifts < 0.5 km/s; spread unchanged at 12.9 km/s; ± 13 budget validated. Pre-frame caches preserved at `results/annular_bootstrap_07c_ishape_oldframe/`. NOTES addendum (d).

D5 Aperture choice — Three apertures considered: $R < R_e/8$ ($=0.288''$), $R < R_e/2$ ($=1.152''$), $R < R_e$ ($=2.305''$). $R < R_e/8$ dropped because it sits inside half the seeing FWHM ($=0.64''$) — only 3 spaxels, and the inner σ would underestimate due to seeing-broadened sampling. $R < R_e$ is the paper headline; $R < R_e/2$ is quoted as a gradient diagnostic.

D6 Equal-N vs equal-width annular binning — For the §7 cross-check only. Equal-N inside $R_{\text{safe}} = 3R_e/4 + 1$ outer flagged bin (`notebooks/07c`) gives balanced bootstrap variance per bin and isolates arc contamination in the outer bin. Equal-width annuli (`notebooks/07b`, 5 bins) shifted σ_e by ~ 10 km/s — at the SPS-systematic level (± 24). Equal-N is the §7 paper choice.

D7 R_e source systematic (mean vs F140W vs F200LP vs Ca H+K + G) — The headline integrates the I-weighted aperture spectrum inside the mean F140W+F200LP masked CoG R_e ($=2.305''$). Three alternative R_e definitions test sensitivity to that choice (Track A masked, $N=500$):

R_e source	R_e ["]	σ_e [km/s]	$\Delta\sigma_e$
mean (paper)	2.305	267.82	—
F140W only	2.168	264.87	−2.96
F200LP only	2.441	275.86	+8.04
Ca H+K + G-band depth	2.866	281.74	+13.92

Spread = 16.9 km/s — at the mask-budget level (± 16) but sub-budget vs the total ± 32 . σ_e rises monotonically with R_e (more outer-bulge spaxels in the I-weighted aperture). The Ca H+K + G-band absorption- depth I-map (rest 3925-3942, 3960-3976, 4297-4313 Å, summed (continuum − flux) per spaxel — see `cahk_g_line_depth_map` in `scripts/final_sigma_e.py`) is the most stellar-only of the four; its larger R_e reflects that the deflector light extends further than the F200LP UV-leaning band suggests. Code: `scripts/final_sigma_e.py` §7 + `cahk_g_line_depth_map`. Display: `notebooks/09 §7b + results/figures/nb09_re_source_systematic.png`. Caches: `results/final_sigma_e_paper/Re_{F140W,F200LP,CaHK}_{sps}_N500.npz`.

E. Mask treatment

Masking vs no-masking — pro/con summary

Track	Pro	Con
Hard mask (w=0.0, headline)	Removes arc contamination; matches literature convention; F200LP mask is purpose-built for this lens	Drops 38 of 184 spaxels inside $R < R_e$ (~21% by count, ~27% by I-weight) → lower S/N; hard boundary may over-clip a few edge spaxels
No mask (w=1.0, sensitivity)	Maximum S/N; no boundary artifacts	Arc adds low-velocity continuum dilution → σ biased low by 15-17 km/s ($\Delta_{\text{mask}} = -16$ km/s at $R < R_e$)
Soft mask (w=0.5)	Smooth middle ground	Super-linear in w (quadratic $c = +7.3$); threshold-dominated bias → first 25% of arc weight already captures 34% of total mask sensitivity. Arbitrary weight choice; not a literature convention
Mask dilation (E6)	—	Inflates σ via noise (dropped spaxels are S/N contributors, not bias contributors). Negative finding — do not dilate

The 16 km/s no-mask Δ is propagated into the error budget as $\sigma_{\text{mask}} = \pm 16$, so the choice is bounded, not assumed.

E1 F200LP arc mask reprojection — `F200LP_mask.fits` (HST 0.04"/pix, arc-tuned) is reprojected to the IFU 0.30"/spax grid via `scipy.ndimage.map_coordinates(order=0)` (nearest- neighbor). 69 of 10000 spaxels (0.69%) flagged. Inside $R < R_e$: 38 of 184 spaxels in the aperture, ~21% — but only ~27% of the I-weight (arc spaxels are mostly off-center).

E2 Hard-mask (w=0.0) headline — track A. E3 No-mask (w=1.0) sensitivity — track B. $\Delta_{\text{mask}} = -15.0$ km/s. E4 Soft-mask single point (w=0.5) — track C. $\Delta_{\text{soft}} = -9.3$ km/s. E5 5-point mask-weight sweep ($w \in \{0, 0.25, 0.5, 0.75, 1\}$) — Per-step σ drops at $R < R_e$: 5.0 → 4.3 → 3.5 → 2.2 km/s. Quadratic $c = +7.3$ (concave-up, super-linear). Threshold-dominated bias: a few heavily-arc-contaminated spaxels carry most of the bias. First 25% of arc weight captures 34% of the total mask sensitivity. Scripts: `scripts/soft_mask_track.py`, `scripts/analyze_mask_weight_sweep.py` Figure: `results/figures/nb09_mask_weight_sweep.png` NOTES addendums (b) and (c).

E6 Mask dilation — Tested whether expanding the F200 mask further reduces σ . Negative finding: dilation inflates σ via noise — the dropped spaxels are S/N contributors, not bias contributors. Memory: `~/claude/.../project_nb07d_overmasking_finding.md`. Notebook: `nb07d`.

E7 Arc-spectrum subtraction sibling — Build an arc-only spectrum from F200-masked spaxels, fit $\alpha \times \text{arc} + \beta \times \text{deflector}$ model, subtract. Result matches §6cum within 0.1 km/s → *residual* arc dilution (the small fraction that leaks through the F200 mask) is sub-dominant at production statistics. The F200 mask already does the heavy lifting. Note: this test compares “masked + arc-subtract” against “masked alone” — it does NOT directly validate the no-mask vs masked Δ mechanism (that’s E8). Notebook: `nb07e`.

E8 Arc-as-sky mechanism test (rigorous test of the no-mask Δ) — Run §6cum at the *no-mask* $R < R_e$ aperture with the bright outer-arc spectrum passed to `ppxf` as a sky template (free-amplitude additive component, NOT convolved with the deflector LOSVD — physically appropriate since the arc is at $z=1.302$, decoupled from the deflector kinematics). At $N=500$, FSPS+EMILES+XSL pooled:

Track	σ_e [km/s]	Δ from σ_A
A: masked headline	267.82	—
B: no-mask	252.82	−15.00
D: no-mask + arc-sky (E8)	274.64	+6.82

Recovery fraction $(\sigma_D - \sigma_B)/(\sigma_A - \sigma_B) = 145\%$ — i.e. arc-sky overshoots the masked headline by ~ 7 km/s. Interpretation:

- Continuum dilution explains the FULL no-mask Δ . There is no detectable kinematic-blend or template-mix mechanism contributing in the *opposite* direction.
- The 7 km/s overshoot is consistent with the known caveat that the outer-arc-mask spaxels ($R > 3R_e/4$) still receive seeing-blurred deflector light at the few-percent level, so subtracting $\alpha \times$ arc oversubtracts a small amount of *real* deflector flux too. This inflates σ slightly.
- The masked headline (Track A, paper number) remains the cleaner measurement because it doesn't have this oversubtraction artefact.
- $\alpha_{\text{arc}} \approx 0.31$ across all three SPS — internally consistent.

Code: `scripts/run_07f_arc_sky.py` (uses `setup_ppxf_inputs_from_spectrum + ppxf sky=` argument). Display: `notebooks/07f_arc_sky_subtract.ipynb`. Caches: `results/arc_sky_07f/{sps}_N500.npz` + `results/arc_sky_07f.npz`. Figures: `results/figures/nb07f_{arc_template,recovery,posteriors}.png`.

F. Centering

F1 5-center sweep — Tested HST_mean (paper choice), F140W only, F200LP only, IFU peak, arc-suppressed IFU peak. σ_e spread = 3.7 km/s. Carried as ± 4 km/s in the error budget. Notes: `NOTES_centering_investigation_2026-04-27.md`.

F2 F140W vs F200LP centroid — $\Delta = 0.36''$ — driven by the F200LP arc dragging that filter's centroid toward the arc. F140W is the cleaner bulge centroid; HST-mean is the robust compromise.

G. Bootstrap and polynomial degree

G1 Wild bootstrap — Hybrid Rademacher sign-flip $\times$ local-residual scaling (75-pix rolling window). For each iteration: residual = (galaxy — bestfit) $\times$ sign, scaled by local σ . Re-fit, record V/σ . Implemented in `scripts/bootstrap_ppxf.py:run_bootstrap_single_degree`.

G2 Parallel bootstrap — `joblib + threadpoolctl` pinning BLAS=1 per worker (otherwise BLAS thread oversubscription pathology — saw 4-hour runtimes). 8 workers gives ~ 6 min for 6 fits at $N=500$. Script: `scripts/bootstrap_ppxf_parallel.py:run_bootstrap_single_degree_parallel`.

G3 Polynomial degree sweep — Degrees 15-29 (15 values) per fit. Diagnostic: σ vs degree should be flat within bootstrap envelope; monotonic trend indicates the polynomial is absorbing LOSVD signal. Plotted at `nb09 §6.5` and saved to `results/figures/nb09_sigma_vs_degree.png`. Saturation confirmed at this degree range — picked from earlier `notebooks/03c` analysis.

G4 $N=50$ vs $N=500$ — $N=50$ smoke and $N=500$ production agree on σ_e to within 1 km/s. $N=50$ is ~ 6 min, $N=500$ is ~ 50 min. Production headline is $N=500$; the smoke is for quick rebuilds.

H. Cross-checks

H1 §7 discrete Gültekin (refreshed 2026-05-01) — `notebooks/07c §7`, recomputed with frame-aware ppxf via `scripts/refresh_07c_gultekin.py`. Annular sum $\sigma_e^2(<R) = \sum F_j (V_j^2 + \sigma_j^2) / \sum F_j$ with arc-filter on outer annulus.

	Pre-frame-fix (Apr 24)	Post-frame-fix (May 1)	Δ
σ_e (arc-filtered)	254.99 −24.2/+28.4	256.17 −13.0/+12.7	+1.18

$\Delta < \pm 5$ km/s frame budget. Errors halved because the V_{sys} split collapsed from ~ 99 km/s (pre-fix) to ~ 8 km/s (post-fix per-SPS V_{sys} : `fsps −11.2`, `emiles −7.4`, `xsl −3.1`), shrinking the V^2 contribution to the Gültekin sum's MC scatter.

H2 §7b flat- σ extrapolation (refreshed 2026-05-01) — `notebooks/07c §7b`, same refresh. Push the §7 sum into the outer flagged bin assuming $\sigma_{\text{outer}} = \sigma$ at the last clean annulus.

	Pre-frame-fix	Post-frame-fix	Δ
σ_e	271 $-33/+35$	274.37 $-16.2/+17.4$	+3.37

Both H1 and H2 remain consistent with the §6cum headline (267.82) at $<1\sigma$. The point of these cross-checks is preserved by the refresh, and the post-fix bands are tighter, strengthening rather than weakening the consistency.

Note: nb07e (E7, arc-spectrum-subtraction sibling) was *not* refreshed because its result is a relative comparison (matches §6cum within 0.1 km/s) — a uniform per-SPS σ shift moves both arms of the comparison together and the relative match is unchanged.

H3 F200 mask sensitivity (independent) — results/annular_bootstrap_07c_nomask/. Same §6cum pipeline with mask off. $\sigma_e(<R_e) = 250.96$ km/s. $\Delta = -16.4$ km/s. Independently confirms the nb09 mask-weight sweep at $w=1.0$.

I. Earlier sigma-discrepancy work (notebooks/05x)

The 21-test diagnostic notebook resolving the $\sigma \approx 301$ km/s (190-spaxel box, NB01) vs $\sigma \approx 210$ km/s (inner circular aperture, NB05) discrepancy. Resolution: σ depends on aperture; the 190-spaxel box is contaminated by the lensed arc. This work motivated the cumulative-I path (Tests 7-13 explored masking variants, Tests 14-16 went radial, Tests 18-19 tried binning approaches). PowerBin (Test 19) was rejected. The headline analysis moved to circular apertures + cumulative I-weighting (nb06 $\rightarrow$ nb07c $\rightarrow$ nb09).

3. File index

Documents (top-level)

File	Purpose
TESTS_AND_DIAGNOSTICS.md	This file — master test index
NOTES_nb09_frame_fix_and_final_sigma_e_2026-04-28.md	Frame fix derivation + addenda (a)–(d)
HANDOVER.md	Apr 27 σ_e analysis snapshot (pre-frame-fix; superseded by nb09)
HANDOFF_Ir_sweep_2026-04-28.md	I(r) sweep design doc (consumed; sweep complete)
NOTES_methodology_2026-04-27.md	Cube provenance + earlier methodology
NOTES_centering_investigation_2026-04-27.md	5-center sweep result + ± 4 km/s budget
CLAUDE.md	Repo conventions, key results summary
README.md	High-level project description

Notebooks (paper-ready / current)

Notebook	Purpose
notebooks/09_final_sigma_e_paper.ipynb	Paper-ready σ_e (headline 268 ± 32 km/s)
notebooks/04_redshift_verification.ipynb	$z = 0.67564$ line-fit
notebooks/02_streamlined_Bagpipes_SED.ipynb	$\log M_\star = 11.33$
notebooks/08_sersic_total_photometry.ipynb	Sersic-total photometry cross-check
notebooks/07c_sigma_e_equalN.ipynb	§6cum + §7 equal-N annular cross-check
notebooks/07e_sigma_e_arc_subtract.ipynb	Arc-subtraction sibling cross-check
notebooks/05x_sigma_discrepancy_diagnostic.ipynb	Tests 1-21 (foundational diagnostics)

Notebooks (history / reference, not paper-driving)

01_*, 02_*, 03_bootstrap_ppxf_errors, 03b_*, 03c_*, 05_*, 06_*, 07_*, 07a_*, 07b_*, 07d_* — kept for the history of the analysis.

Scripts (production)

Script	Purpose
scripts/final_sigma_e.py	nb09 production driver — runs all 3 mask tracks at $N=500$
scripts/build_nb09.py	Generates the paper notebook from the saved artifacts

Script	Purpose
scripts/bootstrap_ppxf.py	Frame-aware ppxf input prep + wild bootstrap
scripts/bootstrap_ppxf_parallel.py	joblib parallel bootstrap (BLAS=1 per worker)
scripts/measure_Re.py	R_e via masked CoG (multi-band, multi-strategy)
scripts/redshift_verify.py	Line-fitting redshift module
scripts/ifu_spectral_resolution.py	LSF audit

Scripts (sweeps and audits)

Script	Purpose
scripts/audit_ppxf_methodology.py	4 orthogonal correctness audits
scripts/sigma_inst_sensitivity.py	DISPCAL $\times$ {0.5..2.0} sweep
scripts/run_isource_shape_sweep.py	10-shape I(r) sweep
scripts/analyze_isource_shape_sweep.py	I-shape sweep analysis + figure
scripts/soft_mask_track.py	Mask-weight runs (CLI --weight)
scripts/analyze_mask_weight_sweep.py	5-point sweep analysis + linearity fit
scripts/regen_s6cum_nomask_diagnostic.py	Centering investigation
scripts/relock_nomask_Re_N500.py	One-off N=500 relock for the no-mask track

Result files (key)

File	Contents
results/final_sigma_e_paper.npz	nb09 headline + per-SPS + 3-track summaries
results/final_sigma_e_paper/	Per-(aperture, SPS, mask_weight) caches
results/sigma_e_ishape_sweep.npz	I-shape sweep summary
results/sigma_e_ishape_sweep_oldframe.npz	Pre-frame-fix I-shape (audit)
results/annular_bootstrap_07c_ishape/	Per-(shape, SPS) I-shape caches (N=500)
results/annular_bootstrap_07c_ishape_oldframe/	Pre-frame-fix I-shape caches (N=250)
results/annular_bootstrap_07c_nomask/	§6cum nomask caches
results/mask_weight_sweep.npz	5-point mask-weight sweep summary
results/ppxf_methodology_audit.npz	4-audit results
results/figures/nb09_*.png	nb09 paper figures
results/figures/nb07c_ishape_sweep.png	I-shape sweep figure

4. Reproduction recipes

conda activate ISMgas

```
# === Headline pipeline (~50 min at N=500) ===
python scripts/final_sigma_e.py --n-bootstrap 500
python scripts/build_nb09.py
jupyter nbconvert --to notebook --execute --inplace notebooks/09_final_sigma_e_paper.ipynb
```

```
# === Methodology audits (~3 min) ===
python scripts/audit_ppxf_methodology.py
python scripts/sigma_inst_sensitivity.py
```

```
# === I-shape sweep (~50 min at N=500) ===
python scripts/run_isource_shape_sweep.py --n-bootstrap 500 --n-jobs 8
python scripts/analyze_isource_shape_sweep.py
```

```
# === Mask-weight sweep (~25 min total) ===
# (w=0 and w=1 are produced by final_sigma_e.py; only need 0.25/0.5/0.75)
python scripts/soft_mask_track.py --weight 0.25
```

```
python scripts/soft_mask_track.py --weight 0.50
python scripts/soft_mask_track.py --weight 0.75
python scripts/analyze_mask_weight_sweep.py
```

```
# === Smoke run (~6 min, useful for rebuilds) ===
python scripts/final_sigma_e.py --n_bootstrap 50
```

Caches are keyed by `N_bootstrap` and `mask_weight` in the filename — re-runs at the same `N` skip cached fits unless `--force` is passed.

5. What's intentionally NOT done

- Re-run nb09 at higher `N` (e.g. `N=2000`): the pooled posterior is already smooth; SPS systematics dominate the residual statistical variance.
- Re-fit Sersic2D F200LP: the `n=0.30` fit is non-physical due to UV-arc domination; the F140W Sersic fit is the trustworthy one. F200LP_Sersic2D is the I-shape outlier *by design* and is excluded from the budget.
- Voronoi binning — Test 18 / nb05x — abandoned; not pursued at `N=500`.
- PowerBin spatial binning — Test 19 / nb05x — rejected (outer-bin $\sigma > 800$ km/s); not pursued.
- Per-spaxel rotation map at production statistics — no rotation detected at the seeing limit; not paper-driving.
- Full nb07a Sersic-I revisit at the post-frame-fix headline — superseded by the I-shape sweep (D3/D4) which covers the Sersic2D shape.

See `~/ .claude/projects/.../memory/MEMORY.md` for compressed cross-session references, including: - `reference_cumulative_vs_annular_sigma_e.md` — §6cum vs §7 design choice - `reference_sps_systematic.md` — per-SPS V_{sys} + frame-fix details - `reference_ppxf_vacair_handling.md` — vac/air conversion methodology - `reference_kcwi_data_properties.md` — KCWI cube + LSF + paper boilerplate - `project_nb09_final_sigma_e.md` — current headline status